

Intrinsic Finite Element Methods for Fourth-Order Elliptic PDEs on Surfaces

Hao Zhou

School of Mathematical Sciences, Peking University

Advisor: Shuonan Wu

CSIAM 2026 · 6.28 · Chengdu

Outline

Introduction

Fourth-Order Elliptic PDEs on Surfaces

Discretization Strategies

What is a Surface PDE?

Surface PDEs: unknowns are defined on a curved space.

closed surface $\partial\Gamma = \emptyset$

planar curved domain $\Omega \subset \mathbb{R}^2$

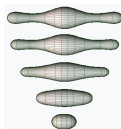
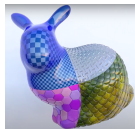
Problems constrained to an interface

Membranes phase fields on biological interfaces.

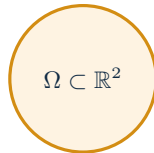
Graphics texture synthesis, mesh processing, visual effects.

Thin shells elastic shells, large bending.

Geometric flows smoothing, shape optimization.



closed surface Γ



curved-boundary plane

What is a Surface PDE?

Surface PDEs: unknowns are defined on a curved space.

Extrinsic: projection from $\mathbb{R}^3$

$$\mathbf{P} = I - \nu \otimes \nu, \quad \nabla_{\Gamma} u = \mathbf{P} \nabla u^e, \quad \operatorname{div}_{\Gamma} X = \operatorname{tr}(\mathbf{P} \nabla X^e).$$

Intrinsic I: exterior calculus

$$\nabla_{\Gamma} u = (du)^{\sharp}, \quad \operatorname{div}_{\Gamma} X = -\delta(X^{\flat}).$$

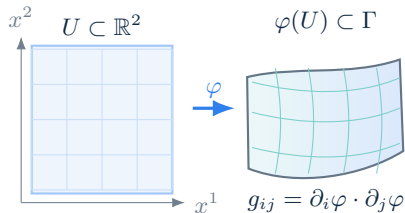
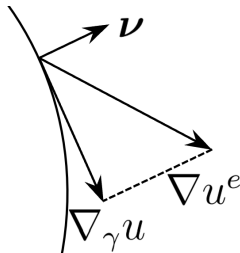
Intrinsic II: metric

$$\nabla_{\Gamma} u = g^{ij} \partial_j u \partial_i, \quad \operatorname{div}_{\Gamma} X = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} X^i).$$

Laplace–Beltrami problem Γ closed, C^2

$$-\Delta_{\Gamma} u := -\operatorname{div}_{\Gamma} \nabla_{\Gamma} u = f, \quad u \in \dot{H}^2(\Gamma),$$

$$(\nabla_{\Gamma} u, \nabla_{\Gamma} v)_{\Gamma} = (f, v)_{\Gamma} \quad \forall v \in \dot{H}^1(\Gamma).$$

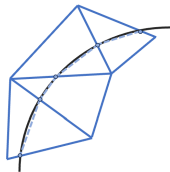


The first step of discretization

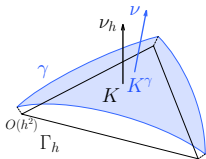
Approximate surfaces



Isoparametric FEM
Surface FEM



Cut FEM
Trace FEM



Surfaces data

explicit $X(\theta, \varphi)$ for S^2

metric $ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2$

level set $\Phi(x) = |x|^2 - 1$

point cloud $\{x_i\}_{i=1}^N \subset S^2$



Pull everything back to a planar PDE?

- Need atlas, partition of unity.
- Quadrature issues.

Use Trace FEM?

- Trace FEM is excellent for moving surfaces.
- But we seek an **intrinsic discretization**.

What is an intrinsic discretization?

Geometric criterion

isometry invariant: **not depend on embedding**

Topological criterion

holes, cohomology FEEC

Analytic criterion

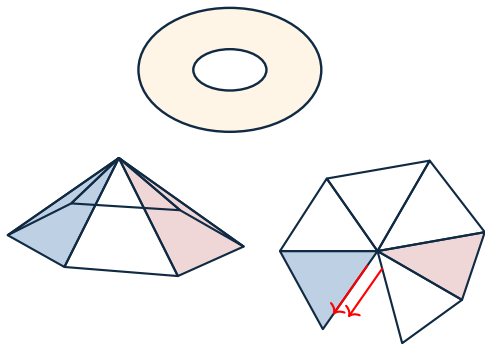
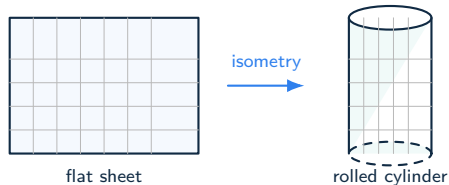
lowest surface regularity assumption

Geometry on piecewise linear surface

no C^0 tangent vector field

- ▶ Extrinsic: elements are tangent to different planes
- ▶ Intrinsic: obstructed by the angle deficit

⇒ no C^1 scalar field

DDGDEC

Fourth-order elliptic PDEs on surfaces

Unified formulation

Find $u \in \mathring{H}^2(\Gamma)$ such that

- ▶ surface biharmonic: $\Delta_\Gamma := \operatorname{div}_\Gamma \nabla_\Gamma$

$$(\Delta_\Gamma u, \Delta_\Gamma v) = (f, v).$$

- ▶ surface Kirchhoff plate: $\nabla_\Gamma^2 = \nabla_\Gamma \nabla_\Gamma$

$$(\nabla_\Gamma^2 u, \nabla_\Gamma^2 v) = (f, v).$$

- ▶ surface Stokes stream function form:
 $H_\Gamma := E_\Gamma(\operatorname{curl}_\Gamma)$

$$(H_\Gamma u, H_\Gamma v) = (f, v).$$

Ignoring lower-order terms.

$$a(u, v) = (\Delta_\Gamma u, \Delta_\Gamma v) - \alpha(K \nabla_\Gamma u, \nabla_\Gamma v).$$

Problem	α
Biharmonic	0
Kirchhoff plate	1
Surface Stokes	2

Goal: A unified discretization without explicit discrete curvature K_h .

- ▶ avoid approximation errors
- ▶ generality as planar case
- ▶ curvature treated implicitly

Distributional FEM

Regge metric

Existing methods

Surface FEM only

Reference	Method	Isom. inv.	Free topol.	Generic no K_h	Optimal conv.	No param.
[larsson2017] [neilan2024]	4th-order primal: C^0 -IP	✓	✓	✓	✓	×
[cai2024]	4th-order primal: reconstruction	×	✓	✓	△	×
[wu2025]	4th-order primal: NZT, Piola dof	×	✓	✓	✓	✓
[wu2026]	4th-order primal: stabilized Morley	✓	✓	✓	✓	✓
[reusken2020]	4th-order mixed: splitting	✓	×	×	✓	✓
[walker2022]	4th-order mixed: HHJ	✓	✓	×	△	✓
[wu2026]	4th-order mixed: decoupled	✓	✓	✓	✓	✓
[bonito2020]	Stokes: div-conforming	✓	✓	✓	✓	×
[hardering2024]	Stokes: penalty normal	×	✓	✓	✓	×
[demlow2024] [demlow2025]	Stokes: Piola dof	×	✓	✓	✓	✓
[nochetto2025]	Stokes: without inf-sup	✓	×	×	✓	✓

✓: yes; ×: no; △: partially / formulation-dependent;

Surface nonconforming FEMs

Surface Morley FEM

- ▶ Same FE space as planar case

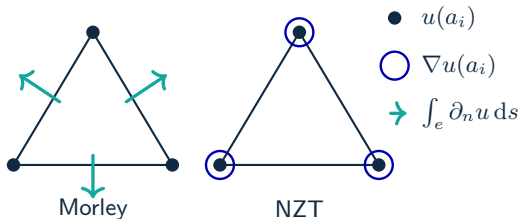
▶

$$a_h(w, v) := (X_{\Gamma_h} w, X_{\Gamma_h} v)_{\Gamma_h} + \sum_{e \in \mathcal{E}_h} h_e^{-3} \int_e [w] [v] ds_h.$$

Surface New Zienkiewicz-Type FEM

- ▶ Nonstandard vector-valued DoFs
- ▶ Stronger continuity
 $\Rightarrow$ No stabilization required in tests

Degree of freedoms



Method	Isom. inv.	Free topol.	Generic no K_h	Optimal conv.	No param.
Morley	✓	✓	✓	✓	✓
NZT	×	✓	✓	✓	✓

Surface nonconforming FEMs

Numerical results for the sNZT element

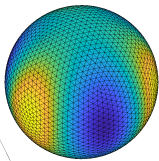


Figure: Numerical approximation of $u = r^{-3}(3x^2y - y^3)$ on the unit sphere.

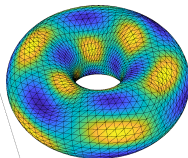


Figure: Numerical approximation of $u = \sin(3\phi) \cos(3\theta + \phi)$ on a torus.

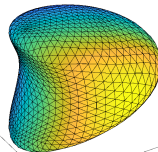


Figure: Numerical approximation of $u = y$ on an implicitly defined surface.

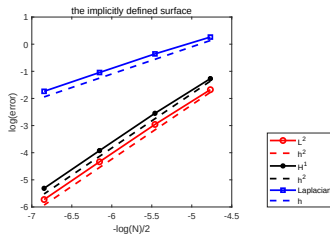
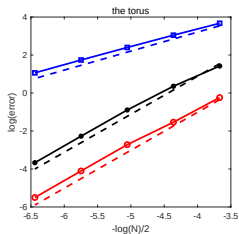
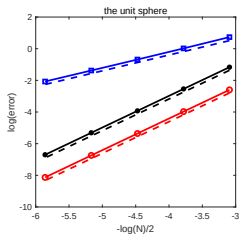


Figure: Error for the surface NZT element using the **stabilization-free** scheme $(\Delta_{\Gamma_h} u_h, \Delta_{\Gamma_h} v_h) = l_h(v_h) \quad \forall v_h \in \tilde{V}_h$, in the three test cases.

Surface mixed FEMs

Problems:

$$(\Delta_{\Gamma}u, \Delta_{\Gamma}v) - \alpha(K \nabla_{\Gamma}u, \nabla_{\Gamma}v) = (f, v).$$

Splitting: $\phi = \Delta_{\Gamma}u$ (Reusken 2018, Elliott 2019)

$$\begin{aligned}(\phi, v) + (\nabla_{\Gamma}u, \nabla_{\Gamma}v) &= 0, \\ (\nabla_{\Gamma}\phi, \nabla_{\Gamma}\eta) + \alpha(K \nabla_{\Gamma}u, \nabla_{\Gamma}\eta) &= -(f, \eta).\end{aligned}$$

HHJ: $\sigma = \nabla_{\Gamma}^2 u$ (Walker 2022)

$$\begin{aligned}(\sigma, \tau) + b(\tau, u) &= 0, \\ b(\sigma, v) + (\alpha - 1)(K \nabla_{\Gamma}u, \nabla_{\Gamma}v) &= (f, v).\end{aligned}$$

second-order variable



K -term



geometric error / no general

Idea: using first-order auxiliary variable

$\nabla_{\Gamma}u$

Surface decoupled schemes

Hodge decomposition: For $g \in L_t^2(\Gamma)$,

$$g = \nabla_{\Gamma} \psi + \operatorname{curl}_{\Gamma} \phi + \xi, \quad \psi, \phi \in \dot{H}^1(\Gamma), \quad \xi \in \mathcal{H}_{\Gamma}.$$

Introduce an integral Lagrange multiplier for $\mathcal{H}_{\Gamma} \Rightarrow$ no topological restriction

Decoupling method

$$\begin{aligned} H^2\text{-problem} &\rightsquigarrow H^1 + \text{Stokes} + H^1 \\ H^m\text{-problem} &\rightsquigarrow \underbrace{H^1 + \cdots + H^1}_{m-1} + \text{Stokes} + \underbrace{H^1 + \cdots + H^1}_{m-1}. \end{aligned}$$

Meanings

- ▶ Bridge between Stokes and fourth-order problems
- ▶ For planar curved domains: boundary DoFs improve accuracy
- ▶ Naturally extends to higher-order elliptic problems

Conclusion

Take-away

- ▶ Intrinsic discretization: embedding-free, topology-independent
- ▶ An intrinsic index: unified treatment of fourth-order problems
- ▶ Primal: stabilized nonconforming FEM; Mixed: decoupled formulation

Ongoing works

- ▶ VEM, HDG, tensor finite elements on surfaces
- ▶ SFEM for geometric quantities
- ▶ Surface viscoelasticity problems

Related papers

- [1] S. Wu and H. Zhou. *A Stabilized Nonconforming Finite Element Method for the Surface Biharmonic Problem*. SINUM, 2025.
- [2] *Stabilized Morley FEM for Surface Stokes in Stream-Function Form: Optimal Convergence via a New Geometric Estimate*. In preparation.
- [3] *A Decoupled Finite Element Method for the surface Kirchhoff Plate Problem*. In preparation.

THANK YOU!

zhouhao23@pku.edu.cn
<https://hao23zh.github.io>